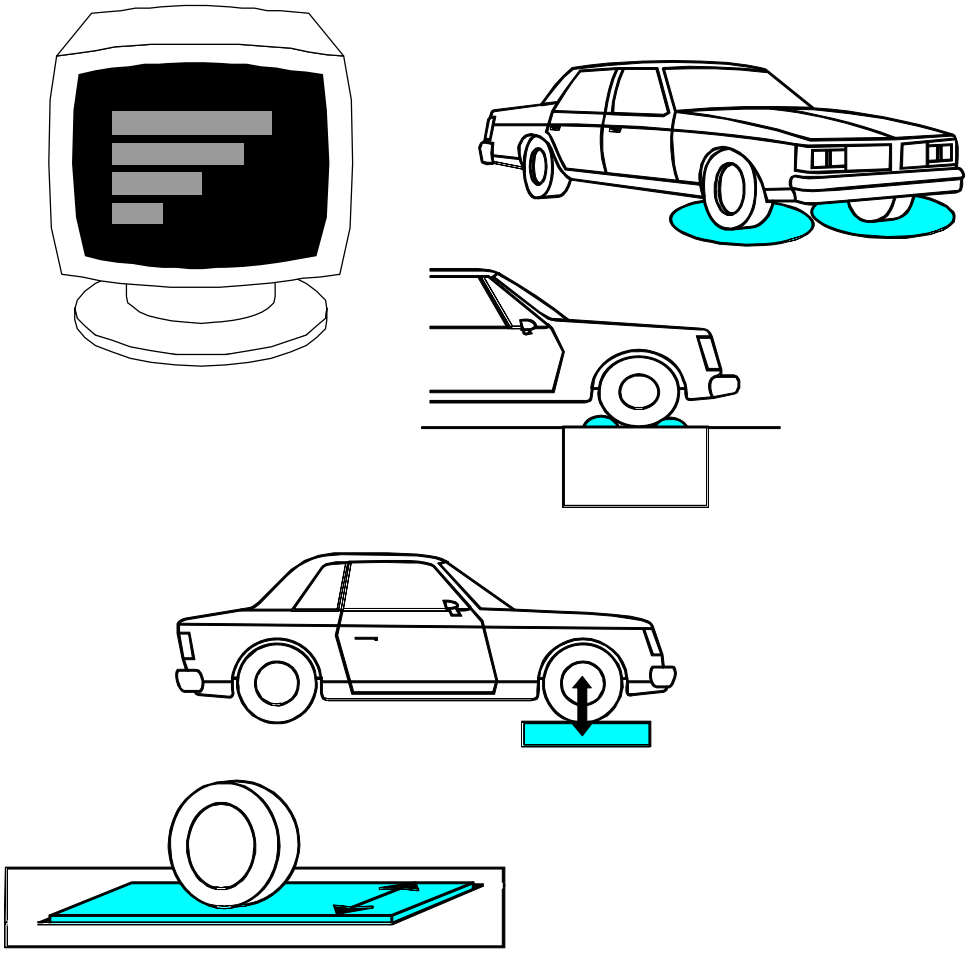
EDP 905 702 002

Operating instructions

SCREEN-TEST-LINE 7000



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1 General information

1.1 Product description - general

The SCREEN-TEST-LINE 7000 may comprise the following devices:
- MICRO-BRAKE 6000
- MICRO-SWING 6200
- MICRO-SIDE-SLIP 6300

Any combination and sequence of the testing equipment is possible (except when the "SUPERAUTOMATIC" program is used - see also Section 3.6).

Auxiliary programs such as the following assure high operating reliability:

- **Checking of all sensors and control units when switched on**
- **Automatic zero-point correction, monitoring for cable rupture and short-circuiting etc. (see also fault messages)**

This description covers all possible options. It is therefore conceivable that the version you have acquired does not perform certain functions described here.

The SCREEN-TEST-LINE 7000 program is for visualising and operating the BEISSBARTH testing line; the PC can be used alone or in addition to the display cabinet as an option.

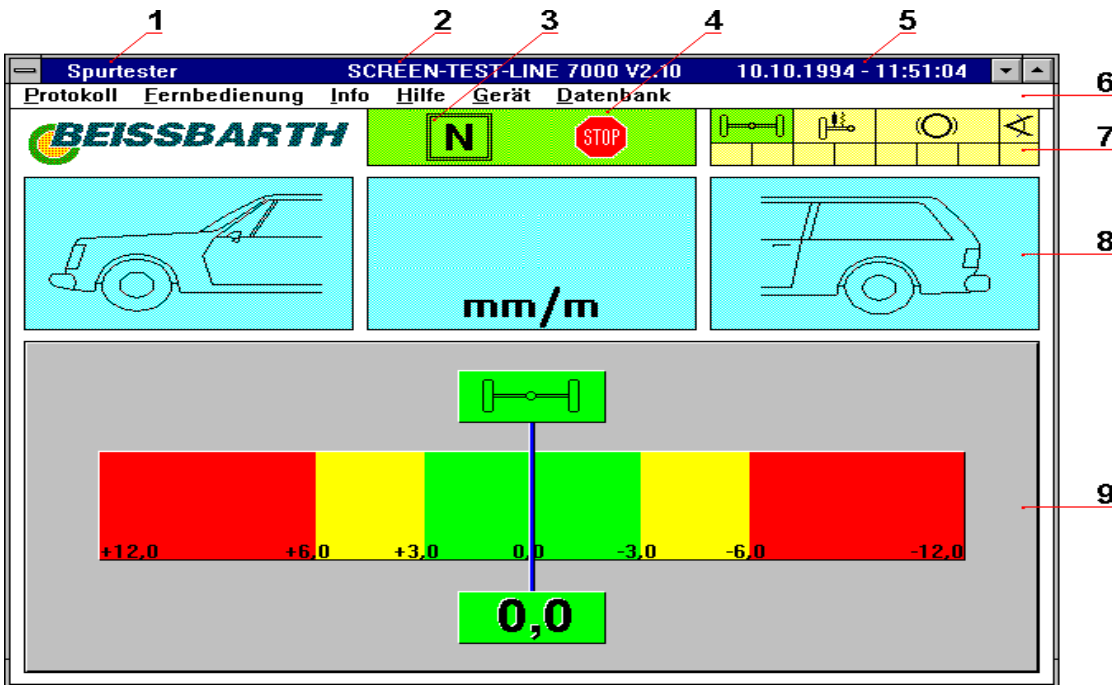
The program communicates with the test-rig's control system via the serial interface and automatically presents testing results in graphic form. Results are expressed as values, graphs and bars.

The remote control field can also be called up in the program; its keys have the same effect as the keys in the infrared control itself.
The PC keyboard also allows inputs of auxiliary information which would not be possible without a PC, such as workshop addresses and customer data which complement the print report.

Visualisation is in Windows 3.1; in other words, the program is conveniently equipped with ultramodern window technology, menu selection and mouse control.

Version 3.5 extends the program with complete storage of measured data in the form of a database. It is consequently possible to switch to the database at any time, store the data and subsequently select them according to various criteria, call them up on the screen and print them out in the form of a report.

The basic layout of the test-rig window looks the same, and is as follows:



1

.. **1** .. Test rig name
2 .. Program title
3 .. Operating mode
4 .. Active remote control button
5 .. Date/time
6 .. Menu/message line
7 .. Sequence of units, indicating the active test rig and storage status of test results (front axle/rear axle/parking brake)
8 .. Stored-value box (front axle/rear axle/parking brake or left/right)
9 .. Measurements/peak-value box

1.1.1 MICRO-BRAKE 6000

This device is a dynamometer for cars and light commercial vehicles. It can be used to check brake systems axle-wise.

1.1.2 MICRO-SWING 6200

This device is a suspension tester for cars, operating according to the EUSAMA principle.

Notes on the measuring principle according to EUSAMA:
The two testing plates are equipped with sensors which allow the weight to be measured.
To assess surface grip, the plate carries out a lifting movement of 6 mm with the aid of an eccentric drive, passing through a frequency band of approx. 23 to 0 Hz in the process.
The resulting sine-wave dynamic wheel load is measured by the electronics and stored. The absolute minimum value calculated in this way is related to the static wheel load and the surface grip calculated. The axle load of the vehicle being tested is then taken into account in the form of a correction factor, to avoid excessively low readings for light vehicles and excessively high readings for heavy ones.

The MICRO-SWING 6200 provides two measured quantities for suspension analysis:

A) Relative surface grip in %:
The following evaluation can be taken as a basis

60 - 100 %	Good
40 - 59 %	Adequate
20 - 39 %	Inadequate
0 - 19 %	Poor

Relative surface grip serves as an indicator of a car's dynamic stability in terms of the shock absorbers' damping effect and the running gear as a whole, such as wheel suspension, wheel bearings, spring struts, tyres etc. Suspension faults can affect relative surface grip as acutely as faulty or worn out shock absorbers.

B) Frequency of resonance in Hz:
The frequency of resonance is an indicator of the "firmness" of the suspension and is influenced in the first instance by the springs in the system. Production vehicles range between approx. 13 and 18 Hz. Values below these are considered soft, and higher values considered sports-standard.

Caution: modifications to the suspension, such as lowering or fitting wide tyres, can affect the results considerably.

How to obtain reliable measurements:

Use the screen-supported technology plus database which we supply. These allow you to store measurements for the following:

1. Vehicles with standard specification
2. Vehicles with optional extras, with modified running gear
3. Customer-specific "reference" vehicles, on which the new or older suspension data can be directly superimposed on the screen.

General assessment:

The difference in relative surface grip between the left and right sides should not exceed 15 %.
The difference in frequency of resonance between both sides should not exceed 3 Hz.

1.1.3 MICRO-SIDE-SLIP 6300

When the vehicle is driven across the toe testing plate, the deviation between the axle's actual geometry and the reference value is displayed.

The following association applies:

No symbol	Toe-in
- (minus)	Toe-out

The following approximate limit values apply:

< ± 4 mm/m	Toe value in order
± 4 to ± 6 mm/m	Toe value borderline
> ± 6 mm/m	Electronic wheel alignment recommended

The limit values may fluctuate due to the varying tolerance ranges for individual vehicle models and varying tyre diameters.

1.2 Technical data

1.2.1 MICRO-BRAKE 6000

Max. drive-over load (per axle)	4 t
Max. testing load (per axle)	2.5 t
Testing speed	5 km/h
Measuring range, braking force	6 kN (per side)
Measuring range, weigher	4 t (per axle)
Roller diameter	210 mm
Motor output	3.7 kW
Gear ratios	20.78
Space between rollers	800 mm
Length of rollers	700 mm
Measuring system	DMS technology
Noise emissions L_{WA}	< 70 dB (A) acc. to DIN 45635 Part 1
Weight, roller set	420 kg
Weight, display cabinet	43 kg
Weight, printer container	28 kg
Weight, support column	23 kg
Enclosure rating, display	IP 54

1.2.2 MICRO-SWING 6200

Max. drive-over load	4 t (per axle)
Max. testing load	2 t (per axle)
Noise emissions L_{wA}	< 70 dB (A) acc. to DIN 45635 Part 1

1.2.3 MICRO-SIDE-SLIP 6300

Max. testing load	2 t (per wheel)
Measuring range	± 12 mm/m
Length x width x installed depth	750 x 440 x 46 mm

2 Instructions for safe operation

2.1 Safety features

2.1.1 MICRO-BRAKE 6000

- If a difference of at least 4 seconds between the left and right sensing roller position occurs (one at top, one at base), the automatic mode is left.
- If braking force does not change for more than 60 seconds, the test rig shuts down the motors. The automatic mode is likewise left.
- If the vehicle's wheels lock when the motors start up, the test rig cuts out immediately.
- During testing of the brakes, the test rig automatically cuts out as soon as one wheel starts to lock.
- The test rig only starts up if pressure is applied to both sensing rollers (not suitable for single-track vehicles).

2.1.2 MICRO-SWING 6200

- Testing is only possible if both vibrating plates are subjected to a load of at least 100 kg each
- Testing is only possible if both vibrating plates are subjected to a load of no more than 1000 kg each
- The automatic mode is left if the vehicle loses contact with the vibrating plates during testing

2.2 Rules for safe operation - general

- The test rigs may only be used in accordance with their designated purpose
- Testing work may only be carried out by trained personnel
- Repair work may only be carried out by trained service personnel
- All electrical installation work must be carried out exclusively by skilled electricians
- No persons or objects may remain within the immediate vicinity of the test rigs during testing work
- Before testing the vehicle, check that the cold tyres are inflated to the correct pressure
- Do not exceed the maximum testing and drive-over loads for the individual test rigs

2.2.1 MICRO-BRAKE 6000

- Apply the brakes slowly, gradually increasing the pressure
- Position the vehicle centrally on the rollers, pointing straight ahead
- Drive slowly onto the test rig

Driving off the test rig:

- Start the vehicle's engine
- Engage 1st gear and depress the clutch
- Switch on the roller set
- Wait until the lock telltales stop flashing
- Release the clutch and drive swiftly off the test rig
- If the vehicle needs to be reversed off the test rig, engage reverse gear now and drive the vehicle off the test rig with a certain amount of momentum, to overcome the mechanical resistance of the rollers

Testing four-wheel-drive vehicles:
On vehicles with driver-selected four-wheel drive, disengage this function before commencing testing work.
On vehicles with permanent four-wheel drive, observe the test specifications for that particular model (see also testing program for four-wheel-drive vehicles).

2.2.2 MICRO-SWING 6200

- Position the vehicle centrally on the testing plates, pointing straight ahead
- The wheels to be tested should not be resting on the frame

2.2.3 MICRO-SIDE-SLIP 6300

- Position the vehicle centrally on the toe plate, pointing straight ahead
- Drive over the toe plate at a constant speed (approx. 5 km/h)

2.3 Impermissible forms of use

2.3.1 MICRO-BRAKE 6000

- Driving in at an angle, with a tyre scraping against the test rig's frame
- Driving onto or across the roller mechanism too quickly
- Actuating the brakes before the lock telltales go out
- Depressing the brakes too abruptly
- Abrupt steering movements when testing the front axle
- Testing single-track vehicles (motorcycles, ...)
- Driving off the test rig without switching on the electric motors (the bearings will otherwise be damaged)

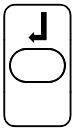
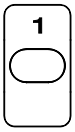
2.3.2 MICRO-SWING 6200

- Entering or leaving the vehicle during testing
- Driving out of the test rig during testing

3 Testing programs

3.1 Regular mode

The regular operating mode of the SCREEN-TEST-LINE 7000 is with the standard testing program, operated by remote control. The "ENTER" key is used to proceed to the next unit. Operation of the individual units is described below.

	Proceed to next unit (sequence can be determined in the configurating routine)
	Double-click the symbol for the unit at the top right of the screen with the PC mouse
	Proceed to next unit with drive-off aid for dynamometer - recommended if the driven axle is located on the roller set.

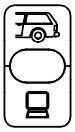
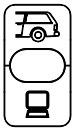
The standard testing procedure for an axle is as follows:

	Change over to toe testing plate
	Drive over toe testing plate and stop on the suspension tester's vibrating plates
	Store toe value
	Change over to suspension tester
	Switch on motors (both wheel slip telltales flash)
	After completing testing, store with remote control for front axle
	Change over to dynamometer
	Drive front axle into roller set
	Switch on motors (both wheel slip telltales flash)
	Gradually increase braking pressure until the wheel slip telltales go out
	Measure any imbalance if necessary
	Apply maximum braking force. Test rig is shut down via wheel slip, sensing rollers or "STOP" button
	Store with the remote control for front axle

3.2 Individual-wheel mode

This operating mode is possible on the MICRO-BRAKE 6000 and MICRO-SWING 6200 and is as the regular mode, except that the wheels on each side are tested individually. Note that the results should be stored after every testing procedure.

Example:

	Testing rear left wheel
	Test completed
	Store rear axle
	Testing rear right wheel
	Test completed
	Store rear axle

The individual-wheel mode on the MICRO-BRAKE 6000 is recommended in particular if the vehicle tends to rear up when braked.

3.3 Automatic mode

This operating mode is possible on the MICRO-BRAKE 6000 and MICRO-SWING 6200. The automatic-mode program is particularly effective for rapid testing, as the vehicle can be driven into the test rig without remote control and the test procedure starts up automatically. The automatic mode is active if both wheel slip telltales (arrows) are lit up when there is no vehicle on the rollers.
If the test rig is set to the automatic mode and measurements need to be stored, the STOP button should be pressed before starting up the motors again. Only then is storage possible.

Important:
If any of the test rig's various safety functions is triggered off or a different unit is selected, the automatic mode shuts down (see also Section 2).

3.4 Imbalance measurement

On the MICRO-BRAKE 6000, the imbalance measurement function can be switched on during measurement (braking forces must be greater than 500 N). It can be repeated as often as desired, and the last 3 measurements are stored (imbalance values are stored by the command to store values for the complete axle).

3.5 Four-wheel drive

The MICRO-BRAKE 6000 includes a four-wheel-drive program as an option. The left and right pair of rollers are driven in opposite directions; only the braking forces of whichever side is rotating forwards are measured. This program is recommended on vehicles with permanent four-wheel drive, as the distortion to the measurements due to superimposed forces is minimised and the differential lock is protected against overloading. Storage is as for the individual-wheel mode.
Always observe the specific notes on testing supplied in the individual vehicle's operating manual.

3.6 "SUPERAUTOMATIC"

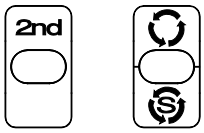
The "SUPERAUTOMATIC" makes it possible to run through a complete test without the remote control; this applies for the following equipment combinations:

- Toe testing plate - vibrating plate set - roller set
- Toe testing plate - roller set
- Toe testing plate - vibrating plate set
- Vibrating plate set - roller set
- Vibrating plate set (individual unit)
- Roller set (individual unit)

To use this function, the components must be installed in the following sequence:

- MICRO-SIDE-SLIP 6300
- MICRO-SWING 6200
- MICRO-BRAKE 6000.

A test cycle with a complete testing line is as follows:

	Activate "SUPERAUTOMATIC" (MICRO-SIDE-SLIP 6300 is activated)
Toe value is stored automatically and the system switches to the MICRO-SWING 6200	Drive front wheels over toe testing plate and halt on vibrating plates
MICRO-SWING 6200 comes on	Front suspension tested
Suspension results are automatically stored and the system switches to the MICRO-BRAKE 6000	Drive onto MICRO-BRAKE 6000 once results for front axle displayed
MICRO-BRAKE 6000 comes on	Service brakes for front axle tested
Imbalance measurement commences	Imbalance measurement (if desired): briefly release brake pedal, then hold constant for a few seconds at a braking force of > 500 N (can be repeated as often as desired; only the last 3 measurements are stored)

When roller set starts again, briefly press brake pedal while "CL" appears in the LCDs (test is then automatically repeated)	Erase braking results if desired
Braking results for front axle are automatically stored and the system switches to the MICRO-SIDE-SLIP 6300 after the vehicle has left the roller set	Drive out of roller set (once "CL" in display has gone out)
Toe value is automatically stored and the system switches to the MICRO-SWING 6200	Drive over toe testing plate with rear wheels and halt on vibrating plates
MICRO-SWING 6200 comes on	Rear suspension tested
Suspension results are automatically stored and the system switches to the MICRO-BRAKE 6000	Drive onto MICRO-BRAKE 6000 once results for rear axle displayed
MICRO-BRAKE 6000 comes on	Service brakes for rear axle tested
Imbalance measurement commences	Imbalance measurement (if desired): briefly release brake pedal, then hold constant for a few seconds at a braking force of > 500 N (can be repeated as often as desired; only the last 3 measurements are stored)
Halt roller set	If locking limit cannot be reached, simply release the brake pedal to interrupt the test
When roller set starts again, briefly press brake pedal while "CL" appears in the LCDs (test is then automatically repeated)	Erase braking results if desired
MICRO-BRAKE 6000 comes on again	Parking brake tested
When roller set starts up, briefly press brake pedal while "CL" appears on the LCDs (test for rear service brake is then automatically repeated)	Erase braking results if desired
Braking results for parking brake are automatically stored and the system switches to the MICRO-SIDE-SLIP 6300 after the vehicle has left the roller set	Drive out of roller set (once "CL" in display has gone out)
Standard printout with concluding evaluation takes place automatically	

The "SUPERAUTOMATIC" function can be interrupted at any time by the "STOP" button. Any values stored until then are retained; further results can be stored or printouts obtained by the regular operating method.

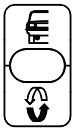
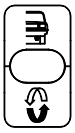
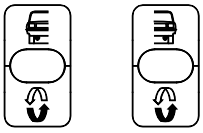
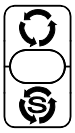
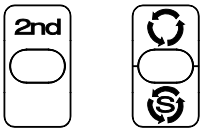
4 Switching on the test rig

The SCREEN-TEST-LINE 7000 is switched on at the mains switch. The switching-on routine is carried out and the printer energised. The test rig must be off-load while this takes place, i.e. no vehicle or other object may be positioned on its mechanical components. Once the screen display switches to the first testing unit, the SCREEN-TEST-LINE 7000 is operational.
The test rig is in the regular operating mode when it is first switched on.
If a system malfunction is identified, it is displayed on the screen and can be acknowledged by pressing the "STOP" button (see also list of fault messages). If this occurs, first of all check whether there is no load on the mechanical components.

5 Operation

5.1 Via infrared remote control

5.1.1 Testing programs

	Individual wheel-testing, left (MB 6000 and MS 6200 only)
	Individual wheel-testing, right (MB 6000 and MS 6200 only)
	Regular mode (press both buttons in any order; MB 6000 and MS 6200 only)
	Automatic mode on (MB 6000 and MS 6200 only)
	"SUPERAUTOMATIC" on

2nd	All-wheel-drive test, left-hand side (MB 6000 only)
2nd	All-wheel-drive test, right-hand side (MB 6000 only)
~	Imbalance measurement (MB 6000 only)
STOP	Interrupt measurement
STOP	Automatic mode off, "SUPERAUTOMATIC" off

5.1.2 Entering gross weight

1000 100 10 ↵	Enter gross weight (MB 6000 only)

This input is displayed in an auxiliary window.
If a weight has already been entered or a weigher is fitted, when the entry mode is called up the actual value is displayed.

Notes:
The weights can be entered before or after the individual tests. Entries can also be made after printouts provided no new measurement has yet been stored.
If a weigher is fitted but a different value is to be used for calculating braking forces, it can be entered as described above (weigher values are overwritten).

5.1.3 Store functions

	Front axle (MB 6000, MS 6200, MSS 6300 only)
	Rear axle (MB 6000, MS 6200, MSS 6300 only)
	Parking brake (MB 6000 only)

5.1.4 Repeat displays

2nd	Gross weight (MB 6000, MS 6200 only)
2nd	Max. values, front axle (MB 6000, MS 6200, MSS 6300 only)
2nd	Max. values, rear axle (MB 6000, MS 6200, MSS 6300 only)
2nd	Max. values, parking brake (MB 6000 only)
2nd	Imbalance (MB 6000 only): If several imbalance measurements are made, measurement 1 is displayed first; if the command is repeated, measurement 2 appears, etc.; unlike the max. value display, only the imbalance values for the axle last tested are shown
2nd	Frequency of resonance (MS 6200 only)

5.1.5 Printout

The desired printout language can be selected in the configuration routine.

	Standard printout with concluding evaluation
2nd	Standard printout
	Graphic printout
2nd	Concluding evaluation

For the national specification "Switzerland" (No.1 in configuration stage 11) the printout for braking forces is reduced by a factor of 0.83
For the national specification "Netherlands" (No. 2 in configuration stage 11), the printout for the braking effect is expressed as braking deceleration in m/s².

5.1.6 Clear functions

CL CA	Erase last measurement
2nd CL CA	Erase complete memory

Stored values can always be overwritten by entering a new value for the same assignment, or storing again for the same axle.
The entire memory is likewise erased if one or more printouts are made and a new measurement is stored.

5.2 Via PC system

In addition to displaying results, additional functions can be called up via the menu line on the screen.

5.2.1 General

The SCREEN-TEST-LINE 7000 can fundamentally be operated as any other application program in WINDOWS ® 3.1 (see also the WINDOWS user manual). The most important control commands are listed below:

F9	Re-write menu line (messages are erased)
F10 or Alt	Activate menu line (desired menu item can be selected with cursor key
Hold Alt pressed + underlined letter in menu line	Desired menu item is called up
Tab or Shift + Tab or cursor keys	Moves the marked remote-control key
Spacebar or Enter	Activates marked remote-control key
Alt + F4	Exit program
ESC	Close auxiliary window
F8	Alter focus (marking) between auxiliary and main window
Tab or Shift + Tab	Select boxes of the auxiliary window, such as switches or input lines
Spacebar or Enter	Activate the marked window switch
Backspace	Correct the input (workshop, report and gross weight window)
Del	Text correction for input lines (workshop and report window)

The desired operating commands can fundamentally also be activated directly with the mouse. To do this, move the mouse to the desired position. Click the left-hand mouse key to carry out the command.

5.2.2 Remote control via screen

With the aid of the menu item Remote, a window showing all remote-control buttons can be opened on the right-hand side of the screen. The SCREEN-TEST-LINE 7000 can be operated as with the regular remote control via these buttons.

5.2.3 Protocol

Once Protocol has been selected, it is possible to choose between the following functions:

Entry...	Input box for customer data is opened (the data then appear on the printout and can also be used for storing results in the database - see also "Database")
Text	Shows the measurements in the standard printout format on the screen
Graphics	Shows the measurements in the graphic printout format on the screen
Olivetti DM 109	A tick must appear before the text if the specified printer is connected up
Language A	A tick must appear before the text if a type A national language is desired; if there is no tick, the languages can be selected from type B (see also the configurating routine)
Exit program Alt+F4	This terminates the visualising program (only necessary if another application program is to be used)

5.2.4 Info

When Info is called up, a window showing the version number, date of completion, copyright details etc. is opened.

5.2.5 Changeover between bars and curves (MB 6000 only)

On the MB 6000, the momentary measurements can be represented graphically in the form of bars and curves.

5.2.6 Changeover to database

After calling up Database, the STL 7000 switches to the database functions.

6 Database

6.1 General information

The database permits convenient administration of all test-rig results obtained and the accompanying customer-specific and model-specific data. It extends the scope of the SCREEN-TEST-LINE 7000 by several important items:

- Storing of test-rig results with customer-specific and vehicle-specific data
- Extensive searching function for loading special customer entries
- Storing reference curves for the suspension tester
- Superimposing a reference curve on actual suspension data
- Making backups of the stored data
- Copying back the backup data

The following tester results from the SCREEN-TEST-LINE 7000 can be stored:

The results from the - Brake tester
- Suspension tester
- Toe tester

and also - Customer-specific data
- Vehicle-specific data

Breakdown of customer-specific and vehicle-specific data:

Customer-specific data
- Name
- First name
- Street
- Postcode
- Town
- Telephone number
- Customer number

Vehicle-specific data
- Registration number
- Vehicle type
- Chassis number
- Distance on clock
- Date of testing
- Time of testing
- Reference curve for suspension tester
- Internal file ref. for results file

6.2 Loading

When Load has been called up, the following functions are available:

Results	Data record can be selected via search mask and processed within the database (e.g. display of results, printout etc.).
Vehicle reference curve (Same vehicle type, different owner)	Only vehicle-specific inputs are taken into account here. When a data record has been selected, the curves for surface grip are transferred from the MS 6200 to the STL 7000, and then superimposed on the current results once the measurements have been stored
Customer reference curve (Same vehicle type, same owner)	Unlike the vehicle reference curve, the customer data can also be used as search criteria. If a data record is selected, the data from the input mask also appear on the printout.

6.2.1 Loading results

This function is used to locate test results already stored in the database. Search keys according to which the data records are selected can be entered in the input mask. When results have been selected, the printout screen is automatically shown. It is now possible to print out the results, etc.

6.2.2 Loading reference curves

Reference curves are called up if you wish to superimpose data stored at an earlier date on the latest suspension test results.
It is particularly pertinent to compare the data for the same vehicle later on in its operating life, because this highlights the gradual ageing process, which is not normally evident to the vehicle's owner.

When reference curves in the database have been selected, the system automatically returns to the visualising mode. The STL screen then shows a car symbol with vehicle type and tyre size, instead of the company logo.
The reference curves remain stored until they are erased (by clicking with the mouse on the car symbol or pressing the spacebar) or new ones are selected.

6.3 Storing

When Store has been called up, the following functions are available:

Results	Data record is stored in the database along with the entries in the input mask
Reference curves	Operated in the same way as storing results, but with the difference that this test is also included in the reference list for suspension results; i.e. if reference curves are loaded, this data record is also accessible.

6.4 Viewing

When View has been called up, the following functions are available (only possible if a data record has already been loaded):

Text	Shows the standard printout with concluding evaluation (is automatically displayed when a data record is selected)
Graphics	Shows the graphic printout for MS 6200 and MB 6000

6.5 Printing

When Print has been called up, the following functions are available (only possible if a data record has already been loaded):

Full printout	Standard printout with concluding evaluation
Text printout	Standard printout
Graphic printout	Graphic printout for MS 6200 and MB 6000
Concluding evaluation	Concluding evaluation only

The printer must be connected up to the PC.

6.6 Updating

When Update is called up, the following functions are available:

Clear	Erases data records (including for specific periods)
Save	Makes backup of data records (including for specific periods)
Return	Data records already backed up are copied back into the database

6.6.1 Clear

Is normally carried out after backing up, if for example the computer's storage capacity is exhausted or very old data records are no longer needed.

If data for a specific period is to be erased, enter "Yes" in the window "Clear for specific period?". The desired period can then be entered in the following time window.

If the above question is answered with "No", the input mask in which the desired boxes are described then appears.

6.6.2 Save

Is normally carried out if for example the computer's storage capacity is exhausted or backup copies are to be made on disk. The latter can be performed very easily by saving for a specific period.

6.6.3 Return

Returning is possible if data that have been backed up and then erased are needed again.

6.7 Info

When Info is called up, a window showing the version number, date of completion, copyright details etc. is opened.

6.8 Changeover to visualising

To leave the database in order to return to the visualising program, select the menu item Visualise.

6.9 Messages on the screen

The process has been interrupted Resume process	<Interrupt> key was pressed during backing up or copying back. Press <Yes> to resume at the point when interrupted, or <No> to interrupt

Identical table entry located Entry was not stored	Message occurs during storing if the data record to be stored is identical to one already in the database. It also occurs if an attempt is made to copy back a backup file which contains exactly the same date record as one already stored in the table. The following must correspond: • Name • First name • Street • Postcode, town • Customer number • Registration number • Vehicle type • Tyre type • Chassis number • Reference curve • Tester • Date of test • Time of test
Error during erasing. The erasing procedure was interrupted.	This error can occur during erasing if an error occurs while erasing an entry in a table, e.g. attempting to erase an entry from an empty database.
Minimum input not satisfied	Occurs if none of the following reference boxes was completed before "Store results" or "Store reference curve". Reference boxes: • Name • First name • Customer name • Registration number • Vehicle type • Tyre type • Chassis number

Too many data records selected - search interrupted	Occurs if the search criteria apply to more data records in the database than can be accommodated in the reference list within the search functions "Search", "Clear", "Store results" or "Store reference curve". The reference list has a capacity of max. 50 data records. **Effects:** Not all data records from the database table which satisfy the search criteria are contained in the reference list. The user can only obtain access to the data records which are contained in the reference list.
Search unsuccessful	May occur during the following functions: "Search" "Clear" "Store results" "Store reference curve" "Load results" "Load reference curve" The database contains no entry which corresponds to the search criteria as defined in the input dialogue.
Insert new disk	May occur during the following function: "Save" If the target disk no longer has sufficient space for storing a further table entry and its results file.
Insufficient capacity on target disk drive. Copy-back interrupted	May occur during the following function: "Return" If an attempt is made to copy back a backup and there is insufficient capacity on the target disk drive to store a further table entry and its results file there. Remedy: erase various files/programs which are no longer needed.

Error in copying results file. This entry is not included.	May occur during the following function: "Return" In the event of an unsuccessful attempt to copy the results file to which the table entry to be copied back belongs. This table entry is then not included in the target table. However, the database completes its process with the next table entry.
Insufficient memory for datasave	This error can always occur if any database function is called up by the SCREEN-TEST-LINE and there is insufficient memory for data exchange.
Data copying unsuccessful	This error can always occur if data are being exchanged between the SCREEN-TEST-LINE and the database. This is always the case when changing from the database to the SCREEN-TEST-LINE or vice-versa. Main cause: The database has not been activated or the SCREEN-TEST-LINE has not been activated and an attempt is made to call them up.
Data record not available	Indicates that a data record which is to be stored is not yet present in the database and can therefore be stored in the database.

6.10 Examples

6.10.1 Storing results

Process	**Comment**
Select Database menu item	Switches to database after testing completed
Select Store menu item	It is now possible to choose whether only the results are to be stored or whether the test should be included in the list of reference data for the MS 6200.

Select Results menu item	The input mask in which the desired boxes are described now appears; if report inputs were already made during STL visualisation, they are also included
Complete input mask	At least 1 reference box must be completed
Confirm inputs	The database now checks whether data records with the same inputs already exist. If this is the case, it is possible to browse through the individual masks. The desired entry should then be confirmed. This version is useful if a customer has already been entered in the database at a previous point in time; it is merely necessary to specify certain search keys; all other entries can then be adopted from the existing input mask, if desired. If there is no data record which corresponds to the inputs made, this is indicated by the message "Data record unavailable".
Confirm input	Input mask is flashed up again and can be corrected if desired
Confirm input	The message "Operation complete" appears; i.e. the data record has been stored in the database. Visualising is then automatically activated again.

6.10.2 Storing reference curves

This is performed in the same way as storing results, with the difference that the box Reference curves is selected in the Store sub-menu.

6.10.3 Loading results

Process	Comments
Select Database menu item	Switches to database
Select Load menu item	It is now possible to choose whether to load results only, or whether reference data for the MS 6200 are to be transferred to the visualising mode
Select Results menu item	The input mask in which the desired boxes are described now appears
Complete input mask	Enter desired search keys
Confirm inputs	The database now checks whether there are data records with the same inputs. If this is the case, it is possible to browse through the individual input masks.
Confirm desired data record	The standard printout of the test that has been searched for is displayed. Further functions such as graphic display, printout etc. are now possible. If the menu item "Visualise" is used to return to the STL, the data record previously selected is closed again.